

Adding and subtracting numerical fractions

While TSM 099 studies Topic 23. Review: Topic 21, Topic 22.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: adding fractions

1. Evaluate and simplify.

$$\frac{17}{12} = \frac{17}{12}$$

2. Evaluate and simplify.

$$\frac{2}{6} = \frac{1}{3}$$

3. Evaluate and simplify.

$$\frac{2}{4} = \frac{1}{2}$$

4. Evaluate and simplify.

$$\frac{20}{6} = \frac{10}{3}$$

5. Evaluate and simplify.

$$\frac{-10}{6} = -\frac{5}{3}$$

6. Evaluate and simplify.

$$\frac{9}{6} = \frac{3}{2}$$

7. Evaluate and simplify.

$$\frac{16}{15} = \frac{16}{15}$$

8. Evaluate and simplify.

$$\frac{5}{8} = \frac{5}{8}$$

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Core: subtracting fractions

9. Evaluate and simplify.

$$\frac{-1}{8} = -\frac{1}{8}$$

10. Evaluate and simplify.

$$\frac{-42}{24} = -\frac{7}{4}$$

11. Evaluate and simplify.

$$\frac{45}{30} = \frac{3}{2}$$

12. Evaluate and simplify.

$$\frac{-30}{12} = -\frac{5}{2}$$

13. Evaluate and simplify.

$$\frac{-16}{8} = -2$$

14. Evaluate and simplify.

$$\frac{3}{12} = \frac{1}{4}$$

15. Evaluate and simplify.

$$\frac{108}{30} = \frac{18}{5}$$

16. Evaluate and simplify.

$$\frac{-16}{15} = -\frac{16}{15}$$

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Core: mixed addition and subtraction

17. Evaluate and simplify.

$$\frac{23}{24} = \frac{23}{24}$$

18. Evaluate and simplify.

$$\frac{31}{12} = \frac{31}{12}$$

19. Evaluate and simplify.

$$\frac{-21}{20} = -\frac{21}{20}$$

20. Evaluate and simplify.

$$\frac{16}{8} = 2$$

21. Evaluate and simplify.

$$\frac{18}{24} = \frac{3}{4}$$

22. Evaluate and simplify.

$$\frac{-36}{12} = -3$$

23. Evaluate and simplify.

$$\frac{7}{10} = \frac{7}{10}$$

24. Evaluate and simplify.

$$\frac{0}{6} = 0$$

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Extension: longer expressions and reasoning

25. Evaluate and simplify. Show intermediate steps.

$$\frac{19}{20}$$

26. Evaluate and simplify. Show intermediate steps.

$$-\frac{29}{12}$$

27. Evaluate and simplify. Show intermediate steps.

$$-\frac{29}{12}$$

28. Evaluate and simplify. Show intermediate steps.

$$-\frac{13}{6}$$

29. Evaluate and simplify. Show intermediate steps.

$$\frac{43}{20}$$

30. Evaluate and simplify. Show intermediate steps.

$$\frac{5}{6}$$

31. Explain the error and give the correct answer.

$$\frac{4}{12} + \frac{3}{12} = \frac{7}{12}$$

Use a common denominator; do not add the denominators.

32. Evaluate. Explain why the sign of the answer makes sense.

$$\frac{4}{10} - \frac{7}{10} = -\frac{3}{10}$$

The amount subtracted is larger than the starting amount.